

IMMUNE CHECKPOINT INHIBITORS

— One-Pager

Release the T-cell brakes · classes, indications & irAE management (NCCN/ASCO) · IM resident reference

MECHANISM — HOW TUMORS HIDE

T cells read mutated "non-self" peptides on **MHC** via the **TCR** and kill cancer cells. Tumors evade this by engaging **immune checkpoints** — normal brakes that prevent autoimmunity:

- Tumor **PD-L1** → T-cell **PD-1** → T-cell "exhaustion"
- Tumor ligands → **CTLA-4** → same brake

ICIs are monoclonal antibodies that **block these brakes** so T cells stay active. High mutational burden = more neoantigens = better response.

DRUG CLASSES

Target	Agents
Anti-PD-1	nivolumab, pembrolizumab, cemiplimab, dostarlimab
Anti-PD-L1	atezolizumab, durvalumab, avelumab
Anti-CTLA-4	ipilimumab, tremelimumab
Anti-LAG-3 2022	relatlimab (+nivolumab = Opdualag; melanoma)

WHEN TO USE — HIGH-MUTATION CANCERS

Best response in mutagen-driven tumors: **melanoma** (UV), **lung** & **bladder** (tobacco). FDA approvals also span RCC, HCC, head & neck, Merkel cell, MSI-high/dMMR tumors, cervical/endometrial, TNBC, gastric/GEJ, mesothelioma, classical Hodgkin lymphoma, urothelial.

irAEs — SIDE-EFFECT MAP

Category	Examples
Most common	Colitis/diarrhea, hepatitis, pneumonitis, rash, thyroiditis, hypophysitis
Irreversible	Myocarditis, type 1 DM, primary adrenal insufficiency
Most deadly	Myocarditis (~40–50% fatal, rare)

Severe irAEs: ~16% anti-PD-1, ~27% anti-CTLA-4, up to **~55% combo**. Onset usually 8–12 wk; can occur any time, even after stopping. Check **TSH/FT4 q4–6 wk**.

MANAGEMENT — 5-STEP FRAMEWORK

1	Rule out other causes (infection/meds) — stool studies, cultures, imaging
2	Stop the checkpoint inhibitor
3	Grade severity (NCCN/ASCO, G1–G4)
4	Prednisone 1–2 mg/kg/day (G2+)
5	No response in a few days → add 2nd agent [infliximab, vedolizumab, IVIG, MMF]

GRADE-BASED RULE (ASCO/NCCN)

Grade	Action
G1	Continue ICI + monitor (except neuro/some heme); topical/supportive
G2	Hold ICI; steroids; resume when ≤G1
G3	Hold; high-dose steroid, taper ≥4–6 wk; 2nd agent if refractory 48–72 h
G4	Permanently d/c (except endocrinopathy controlled on replacement)

Do / AVOID: DO exclude infection before steroids; DO escalate myocarditis urgently. AVOID treating ICI diarrhea as colitis before stool studies; AVOID permanent d/c for a controlled endocrinopathy on hormone replacement.

Adapted from TeachIM.org "Immune Checkpoint Inhibitors" (Longino, Geiger, Schenk; exec. ed. Fainstad; Nov 2021) · CC BY-NC 4.0 · educational use only, not medical advice · oncology/irAE guidance evolves — verify against current NCCN/ASCO & have clinician-reviewed before teaching · optimized June 2026.